

C: Celestial Veins

Source file name: `celestial.c`, `celestial.cpp`, `celestial.java`, or `celestial.py`

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Eons ago, in a world bathed in perpetual night, a civilization of astronomers and poets --the Luminari-- flourished. Without instruments to plumb the depths of the cosmos, their understanding of the universe rested on what the eye could trace across the firmament. Thus they mapped the heavens as they appeared: a great celestial canvas projected onto a two-dimensional Cartesian plane. All their sacred theories, and every Euclidean distance they computed, were grounded in this planar projection.

They held that stars forged bonds of proximity and reciprocity, governed by a single rule: the *Law of Shared Skies*. According to this law, two distinct stars A and B form a true *bond* iff each regards the other as a close companion. Concretely, each star must lie within the other's *trusted circle*, defined as its K nearest distinct neighbors (by Euclidean distance in the planar projection). Ties at equal distance are broken by the *chronicle index*: the star that appears earlier in the ancient charts (input order) takes precedence. Any relationship that is not mutual is dismissed as a fleeting alignment, not a bond.

For the Luminari, a *constellation* is any group of at least two stars in which every pair is linked by an unbroken chain of bonds (i.e., a connected component of the mutual-bond graph with size greater than one). The pattern of bonds determines the constellation's essence, and each constellation is classified into exactly one of the following five sacred forms, evaluated in this order of priority:

Stellar Cluster: A group of stars where every star is bonded to all others, thus forming the maximum possible number of bonds for its size.

Cosmic Ouroboros: A closed ring of stars where the number of bonds is exactly equal to the number of stars.

Dragon's Claw: A formation where a single central star acts as a nexus, holding together the rest of the stars in the group, which possess no bonds among themselves.

Traveler's Path: A simple chain of stars that extends between two distinct endpoints, featuring no cycles or branches.

Amorphous Nebula: Any other formation of one or more interconnected stars that does not fit the patterns above. Additionally, any star that remains completely isolated is also counted in this category.

In a forgotten chronicle, the Luminari describe a celestial region they called "The Ring and the Swarm." With observation parameter $K = 2$, their map listed eight stars in the order recorded by the chronicles.

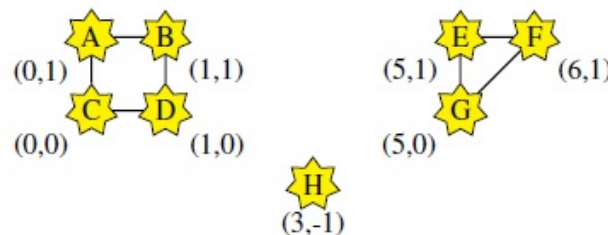


Figure 1: The Ring and the Swarm with adjusted labels

Their analysis, following the sacred classification rules, revealed three formations: one Cosmic Ouroboros (the Ring formed by stars A , B , C , and D), one Stellar Cluster (the Swarm formed by E , F , and G), and

one Amorphous Nebula (the isolated star H). No formations matched the Traveler's Path or Dragon's Claw patterns.

Your task is to apply the Luminari's ancient laws to each star map and report how many formations of each sacred type are present.

Input

The input consists of multiple star maps. Each map begins with a line containing two integers N ($1 \leq N \leq 1\,500$) and K ($1 \leq K \leq 5$). Then follow N lines; each line contains two integers X and Y ($-10^9 \leq X, Y \leq 10^9$) giving the coordinates of a star. The order of appearance defines the chronicle index: the first star listed has the lowest index, the second the next, and so on. The input terminates with a case where $N = 0$ and $K = 0$.

The input must be read from standard input.

Output

For each map, print one line with five space-separated integers: the counts of Traveler's Path, Cosmic Ouroboros, Stellar Cluster, Dragon's Claw, and Amorphous Nebula, in this exact order.

The output must be written to standard output.

Sample Input	Sample Output
8 2	0 1 1 0 1
0 1	1 0 1 0 0
1 1	
0 0	
1 0	
5 1	
6 1	
5 0	
3 -1	
7 2	
0 0	
1 0	
0 1	
100 0	
101 0	
102 0	
103 0	
0 0	